Direct and alternating electrical signals

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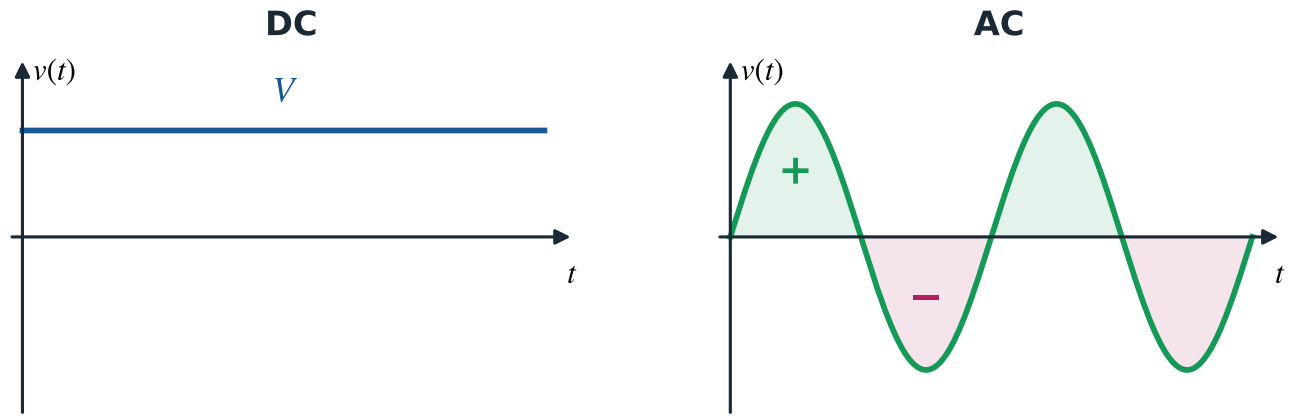
The electrical signal

An **electrical signal** is an electrical quantity, **voltage** or **current**, that carries information or energy and may vary over time.

A naming convention is used to tell values apart:

- **lowercase letter** → instantaneous value, i.e. the value at a given instant: $v(t)$, $i(t)$;
- **uppercase letter** → constant or characteristic value: V, I, V_M, I_M.

Direct and alternating signals



Direct signal (DC)

A **direct** signal (*DC – Direct Current*) keeps **the same value and the same direction** over time.

- On the voltage–time graph it is a **horizontal line**.
- It is written as V (voltage) and I (current).
- Examples: a cell, a car battery (12 V), a bench power supply, a USB port (5 V).

Alternating signal (AC)

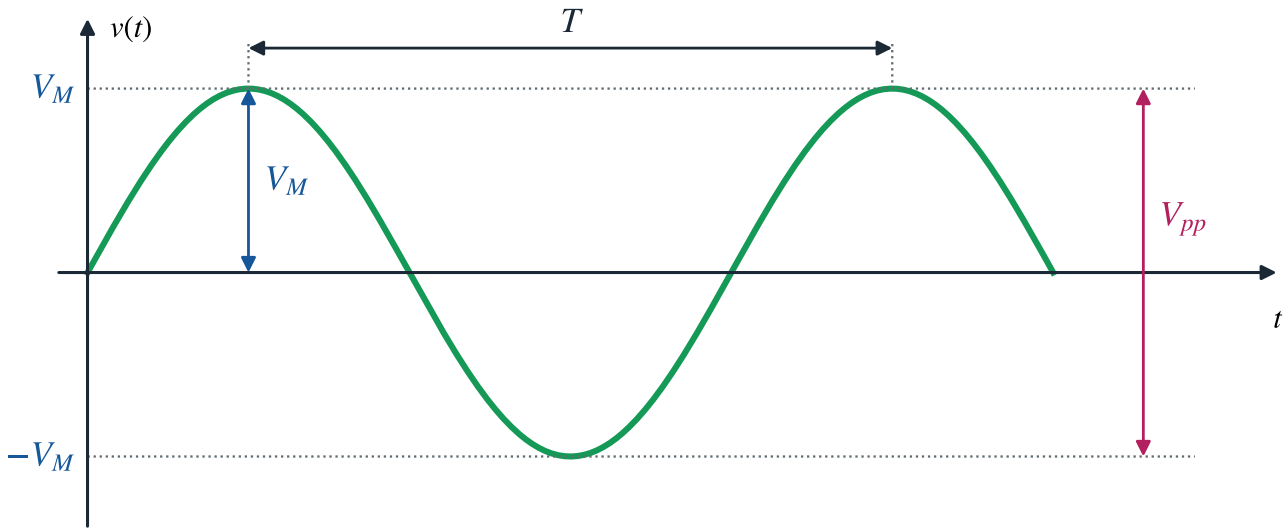
An **alternating** signal (*AC – Alternating Current*) **periodically changes value and direction**: it repeats itself identically over time and takes both positive and negative values.

- Over one period its **average value is zero**: the area of the positive half-wave equals that of the negative half-wave.
- Typical example: the domestic **mains supply** (230 V, 50 Hz in Italy and Europe).

The sine wave

The most important alternating waveform is the **sine wave**. The instantaneous voltage and current are written as:

> **Voltage:** $v(t) = V_M \cdot \sin(\omega\, t + \varphi)$
>
> **Current:** $i(t) = I_M \cdot \sin(\omega\, t + \varphi)$



Symbol	Quantity	Unit	Meaning
V_M, I_M	amplitude (maximum or peak value)	V, A	the highest value reached by the sine wave
V_{pp}	peak-to-peak value	V	distance between maximum and minimum: $V_{pp} = 2 \cdot V_M$
T	period	s (seconds)	time taken to complete one full oscillation
f	frequency	Hz (hertz)	number of complete oscillations per second
ω	angular frequency (omega)	rad/s	angular speed at which the phasor rotates
φ	initial phase (phi)	rad or °	position of the sine wave at $t = 0$
$\omega\, t + \varphi$	phase (instantaneous angle)	rad	angle reached at instant t

Key formulas

> **Frequency and period:** $f = 1 / T \iff T = 1 / f$
>
> **Angular frequency:** $\omega = 2\pi \cdot f = 2\pi / T$
>
> **Peak-to-peak value:** $V_{pp} = 2 \cdot V_M$

Frequency and period are **inversely proportional**: if the frequency doubles, the period halves.

Useful multiples and submultiples: 1 kHz = 10^3 Hz · 1 MHz = 10^6 Hz · 1 ms = 10^{-3} s · 1 µs = 10^{-6} s

Angles in radians

In electrical engineering angles are usually measured in **radians** (rad). A full turn is **2π rad**, i.e. 360°.

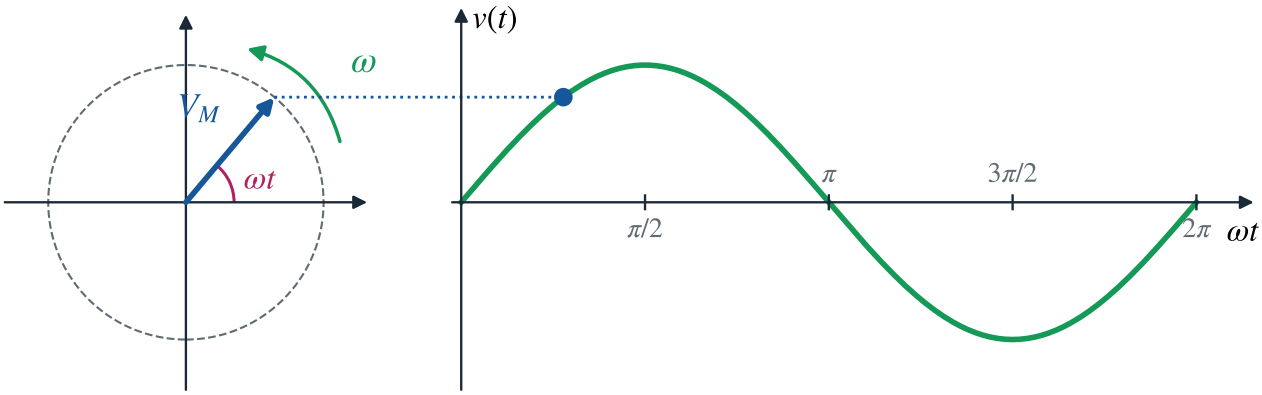
> **Degrees to radians:** $\alpha_{rad} = \alpha° \cdot \pi / 180$
>
> **Radians to degrees:** $\alpha° = \alpha_{rad} \cdot 180 / \pi$

Degrees	**0°**	**30°**	**45°**	**60°**	**90°**	**180°**	**270°**	**360°**
Radians	0	π/6	π/4	π/3	π/2	π	3π/2	2π

This is why angular frequency is also called **angular speed**: in one period T the phasor makes one full turn, i.e. it sweeps 2π rad, so $\omega = 2\pi / T$.

Phasor representation

Drawing and adding sine waves is awkward. For this reason each sine wave is represented by a **phasor**: a **vector rotating anticlockwise** around the origin.



- The **length** of the phasor is the **amplitude** V_M.
- The **rotation speed** is the **angular frequency** ω.
- The **initial angle**, measured anticlockwise from the horizontal axis, is the **initial phase** φ.
- The **projection onto the vertical axis** of the rotating phasor traces, instant by instant, the **sine wave** $v(t)$.

A phasor is written as an uppercase letter with a bar, $\bar{\mathbf{V}}$, in polar form:

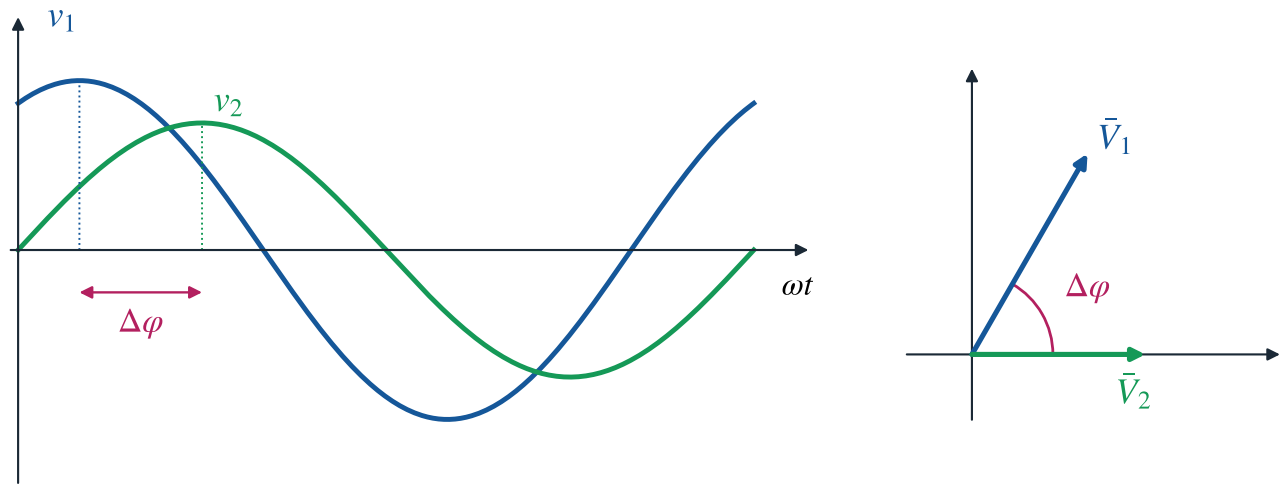
> **Phasor:** $\bar{V} = V_M \angle \varphi$

When all quantities have the **same frequency**, the phasors rotate together and keep the same angles between them. They can therefore be drawn **stationary**, in their position at $t = 0$: this is the **phasor diagram**.

Phase and phase shift

The **initial phase** φ tells where the sine wave starts at $t = 0$. Comparing two sine waves **with the same frequency** gives the **phase shift**:

> **Phase shift:** $\Delta\varphi = \varphi_1 - \varphi_2$



In the phasor diagram the phase shift is simply the **angle between the two phasors**.

- **in phase** ($\Delta\varphi = 0$): they reach their maximum at the same instant;
- **leading** ($\Delta\varphi > 0$): the first reaches its maximum before the second (in the figure v_1 leads v_2);
- **lagging** ($\Delta\varphi < 0$): the first reaches its maximum after the second;
- **in phase opposition** ($\Delta\varphi = \pi$ rad $= 180°$): when one is at its maximum, the other is at its minimum;
- **in quadrature** ($\Delta\varphi = \pi/2$ rad $= 90°$).

The phase shift can also be expressed as a time delay: $\Delta t = \Delta\varphi / \omega$.

Reading a signal on the oscilloscope

The **oscilloscope** plots voltage against time. The screen is divided by a grid into **divisions** (usually 10 horizontally and 8 vertically), and each division is further split into **5 parts**, the ticks on the central axes. One part is therefore **1/5 of a division = 0.2 div**.

To turn divisions into volts and seconds you need to know the two oscilloscope settings:

- **V/div** (volts per division, vertical control): how many volts **one vertical division** is worth. Each channel has its own V/div.
- **Time/div** (time per division, horizontal control): how much time **one horizontal division** is worth. It is the same for all channels.

Counting divisions and parts

First count the **whole divisions**, then the remaining **parts**, and add them:

> **Divisions read:** n = whole divisions + parts · 0.2

Parts	1	2	3	4	5
Divisions	0.2	0.4	0.6	0.8	1

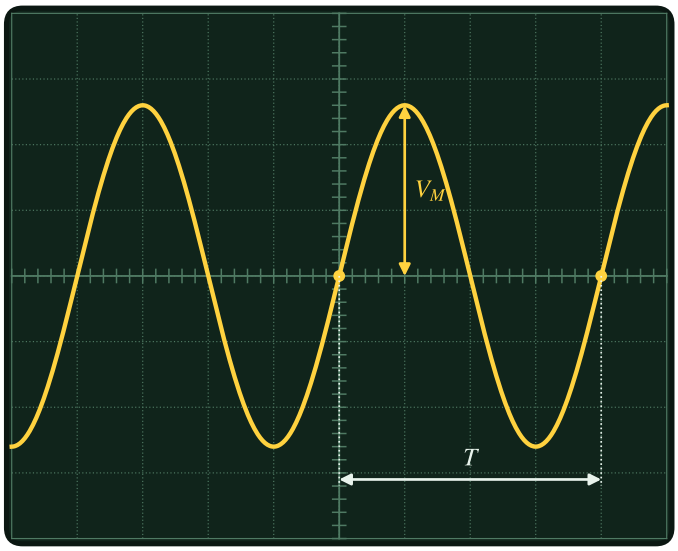
For example 2 divisions and 3 parts give 2 + 3 · 0.2 = **2.6 div**.

Voltage, period and frequency

The divisions read are turned into values by multiplying by the oscilloscope setting:

> **Voltage:** $V = n_{\text{vertical}} \cdot \text{V/div}$
>
> **Time:** $t = n_{\text{horizontal}} \cdot \text{Time/div}$
>
> **Frequency:** $f = 1 / T$

- **Amplitude V_M:** count the vertical divisions from the central axis (zero) up to the peak. It is more accurate to measure the peak-to-peak value V_{pp}, from peak to trough, and divide by 2: $V_M = V_{pp} / 2$.
- **Period T:** count the horizontal divisions between two identical consecutive points of the sine wave, for example two rising zero crossings.
- **Frequency f:** it is found from the period, $f = 1 / T$.



Example. Settings: **2 V/div** and **1 ms/div**.

- **amplitude:** the peak is at 2 divisions and 3 parts = 2.6 div → $V_M = 2.6 \cdot 2$ V = **5.2 V** (V_{pp} = 10.4 V)
- **period:** there are 4 div between two rising zero crossings → $T = 4 \cdot 1$ ms = **4 ms**
- **frequency:** $f = 1 / T = 1 / 0.004$ s = **250 Hz**

- **angular frequency:** $\omega = 2\pi \cdot f = 2\pi \cdot 250 \approx$ **1571 rad/s**
- the sine wave crosses zero rising at the centre of the screen ($t = 0$), so $\varphi = 0$ and **v(t) = 5.2 · sin(1571 t) V**

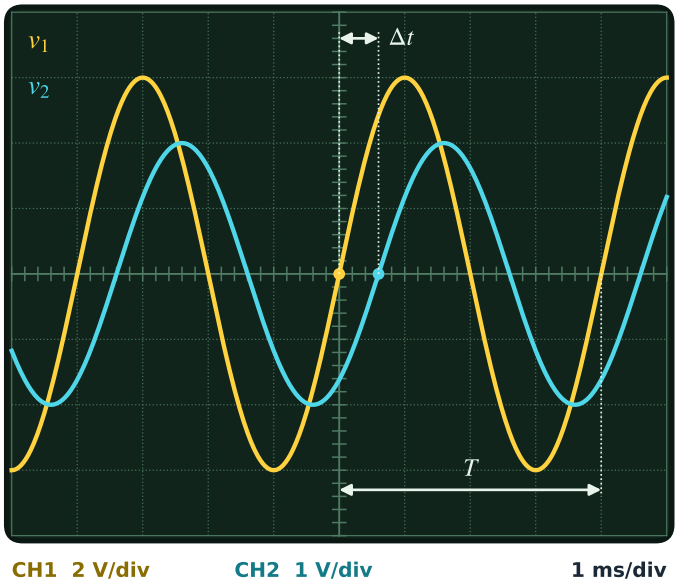
Phase shift between two signals

With two signals of the **same frequency** shown on the two channels:

- **1.** measure the **period *T***, which is the same for both signals;
- **2.** measure the **horizontal distance Δ*t*** between two corresponding points, for example the two rising zero crossings;
- **3.** divide Δt by the period and multiply by 360°.

> **Phase shift:** $\Delta\varphi = (\Delta t / T) \cdot 360°$
> **In radians:** $\Delta\varphi = (\Delta t / T) \cdot 2\pi$

The signal that crosses zero rising first, i.e. the one **further left**, is **leading**; the other one is **lagging**.



Example. Settings: CH1 **2 V/div**, CH2 **1 V/div**, **1 ms/div**.

- **amplitudes:** v_1 reaches 3 div → 3 · 2 V = **6 V**; v_2 reaches 2 div → 2 · 1 V = **2 V** (careful: the two channels have different V/div)
- **period:** T = 4 div · 1 ms = **4 ms**, so f = **250 Hz**
- **time difference:** Δt = 3 parts = 0.6 div → 0.6 · 1 ms = **0.6 ms**
- **phase shift:** $\Delta\varphi = (0.6 / 4) \cdot 360° =$ **54°** (≈ 0.94 rad): v_1 **leads** v_2 by 54°

Tip: choose V/div and Time/div so that the signal fills most of the screen height and one or two periods are visible. This makes the readings more accurate.

Example: the mains supply

For the domestic mains $f = 50$ Hz:

- **period:** $T = 1 / f = 1 / 50 = 0.02$ s = **20 ms**
- **angular frequency:** $\omega = 2\pi \cdot f = 2\pi \cdot 50 \approx$ **314 rad/s**
- **amplitude:** 230 V is the *RMS value*; the maximum value is $V_M = 230 \cdot \sqrt{2} \approx$ **325 V**, so $V_{pp} \approx 650$ V
- **expression:** $v(t) \approx 325 \cdot \sin(314\ t)$ V